The empirical recurrence for OEIS A203916, proved by transfer matrix

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Abstract

OEIS A203916 counts $(n+2)\times 3$ arrays over $\{0,\dots,1\}$ subject to a condition imposed on every 3×3 subblock, and counted UP TO RELABELLING of the alphabet; it carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is equality patterns, which is not a walk count — but it is a fixed rational combination of walk counts, one for each alphabet size, and that combination is again C-finite. Whether the conjectured recurrence annihilates it is then decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A203916 is "Number of (n+2) X 3 0..1 arrays with every 3 X 3 subblock having equal diagonal elements or equal antidiagonal elements, and new values 0..1 introduced in row major order.". It has offset 1 and begins

$$112, 392, 1400, 5000, 17800, 63368, 225704, 803912, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 3*a(n-1) +6*a(n-3) +4*a(n-4).

As of the "Last modified" line on the live entry (Jun 05 2018 14:35:37, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Patterns are a combination of counts

The clause "new values 0..1 introduced in row major order" says that the arrays counted are the canonical representatives of their EQUALITY PATTERNS: two arrays that differ only by a permutation of the alphabet are one object. Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible ARRAYS over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i \;=\; \sum_j N_j\, i(i-1)\cdots(i-j+1),$$

a triangular system with nonzero diagonal, hence invertible over $\mathbb{Q}$. The entry counts $a = N_1 + \cdots + N_2$, so

$$a \;=\; \big(\mathbf{1}^\top F^{-1}\big)L,$$

a FIXED rational combination of the L_i, where F is the matrix of falling factorials. Each L_i is a walk count on its own transfer graph, so a is a walk count on the disjoint union of those graphs with the combination's coefficients placed in the starting vector; the coefficients are rational and some are negative, which costs nothing in what follows.

This step needs the condition itself to be unchanged by permuting the alphabet, and it is: the condition below is stated through equality between entries and never names a particular value or compares sizes.

3 Three consecutive lines

The entry's condition constrains every 3×3 subblock, which spans three consecutive rows and three consecutive columns. Writing an array as a sequence of rows $u_1, u_2, \dots$ over $\{0, \dots, 1\}^3$, the condition

$$\text{the diagonal or the antidiagonal of } g \text{ is constant}$$

involves only three consecutive rows. Take as vertices the ordered PAIRS (r, s) of rows and put an edge $(r, s) \to (s, t)$ exactly when the condition holds for the triple (r, s, t) at every admissible column; an array with L rows is a walk of length $L-2$. Doing that for each alphabet size $1, \dots, 2$ and combining as above gives

$$a(n) \;=\; \frac{1}{2}\,\iota^\top M^{\,n}\tau$$

on $S = 64$ vertices in total, the exponent fixed by the entry's own shape and checked against its published terms. States lying on no walk from ι to τ are removed first; that changes no value and only shrinks the computation.

4 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ satisfy the monic linear recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. ☐

Theorem 2.

$$a(n) \;=\; 3\,a(n-1) + 6\,a(n-3) + 4\,a(n-4)$$

for every $n > 3$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly. Those integers vanish from the index corresponding to $n = 3 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 3$. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. That check earned its place here. The neighbour offsets are named in the array's own frame — horizontal means along a row — but when the entry fixes the first dimension, as in "$2 \times n$", the walk runs along columns and the two components of every offset swap. Sets such as king-move, or horizontal together with vertical, are unchanged by that swap and hide the error completely; a set such as horizontal, diagonal and antidiagonal is not, and the counts came out as those of a different member of the same family until the offsets were transposed.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A203916.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.